# Lecture 27 (Game Theory Introduction)

May 3, 2026

The lectures from this last week of the class are an application of LPs. These lectures are optional, but are interesting.

## Terminology and Definitions

**Game Theory** is the mathematical study of strategic interactions. Despite the name "game theory", mathematical games are not always the same as "games" in the usual sense. Some examples of mathematical games are given below:

1. Chess
2. Checkers
3. Poker
4. Rock Paper Scissors
5. Stock Trading
6. Strategic Voting
7. Football*
8. Basketball*

The games with an * are complicated. Some aspects of these games (such as play calling) are mathematical games. Other details like running to score a touchdown are more about athletic ability than strategy, so they aren't considered mathematical games. We will talk more about what makes a game a mathematical game with some additional definitions. There are a few considerations to make with mathematical games.

### Number of Players

Historically, game theory mainly studies 2 player games. Games with more than 2 players are much more complicated to study. In addition, some mathematicians have studied 1 player games (like solitaire) and 0 player games (these led to some ideas within cellular automata). From the examples above, we have chess, checkers, and rock paper scissors as clear 2 player games. Poker can be played with two players or with many players. The number of players changes the strategy. Stock trading and strategic voting involve many players. These types of games have led to the field of **Social Choice Theory**. Football and basketball are complicated. If each team is viewed as a player and play calling is set with the goal of maximizing the total team score, then these games can be viewed as two player games. If we focus on each player's individual actions, then each of these games can be viewed as multiple player games. For convenience with two player games, we will refer to the two players are Alice and Bob.

### Simultaneous vs Sequential

A game is called a **simultaneous** game if each player takes an action simultaneously. Games such as rock-paper-scissors and football (if viewed from a play calling perspective) are simultaneous games. Games where players take turns making actions are called **sequential** games. These include chess, checkers, and poker. For today we will focus on simultaneous games. Next lecture we will talk more about sequential games.

## Information

A game may be a game of **complete information** or **incomplete information**. Chess and Checkers are games of complete information since each game has no element of chance or uncertainty. Each player also has access to the same information as the opponents. Games like rock paper scissors and poker consist of incomplete information. In rock paper scissors, each player knows what their own move is. If Alice knows what Bob intends to play in rock paper scissors, then the game fundamentally changes. With poker, if Alice knew the other player's cards, the game will change. Even so, there is still the additional incomplete information about what the remaining cards to be dealt are. Games with partial information can be quite interesting to study and understanding what information is known/unknown can change the analysis of the game.

## Strategies

The action or sequence of actions a player determines to take is called a **strategy**. A **pure strategy** is a strategy that involves no element of chance. For example, a chess player may always opt to move their d4 pawn forward two squares at the start of every game. A **mixed strategy** is a strategy that assigning a probability to each pure strategy in a collection of pure strategies. A pure strategy is chosen at random with a probability proportional to the assigned probability. For example, a rock paper scissors player may opt to select rock, paper, or scissors each with probability $\frac{1}{3}$. In rock paper scissors, if Alice always picks rock, Bob is always incentivized to select paper. If Alice selects one of the options randomly, it is much harder for Bob to win. In many games a mixed strategy can be better.

## Payoff

When each player selects a strategy, the **payoff** for each player is given as the amount of "points" given for the outcome of the game. Here, I am being a little bit informal. These "points" could be dollars in poker, actual points (like in football), or could be number of wins (as we see in games like chess). In economics, this might be some measure of utility. For rock paper scissors, we may consider what happens if Alice selects rock and Bob selects paper. The payoff for each player is given in an ordered pair:

$$(\text{Payoff for Alice}, \text{Payoff for Bob}) = (-1, 1)$$

Here, the "points" are the number of games that would be won for each player given they each chose their respective strategies. We can summarize all of the payoffs for each player in a table (such as the one below). Each row will represent the strategy selected by Alice and each column will represent the strategy selected by Bob.

| Payoff | Rock | Paper | Scissors |
|---|---|---|---|
| Rock | $(0, 0)$ | $(-1, 1)$ | $(1, -1)$ |
| Paper | $(1, -1)$ | $(0, 0)$ | $(-1, 1)$ |
| Scissors | $(-1, 1)$ | $(1, -1)$ | $(0, 0)$ |

## Optimal Strategies

In game theory, both players are usually assumed to be rational and to act in their own best interest. If all players make this assumption, then each player has a strategy that will such that changing strategies will worsen their position (assuming the opponent also is playing with perfect play). This strategy is called an **optimal strategy**. These strategies may be pure strategies or mixed strategies.

## Zero Sum Games

A game is a **zero sum game** if the sum of each player's payoff is constant for all pairs of strategies. For example, in rock paper scissors, the sum for each pair will be zero. Another example is if two companies are competing for marker share in a duopoly. If the payoff is the percentage of the market share each company has, then the sum will always add to 100. In games where the sum is non-zero, you can always transform the game to an equivalent game where the sum is 0 (by subtracting the sum from each payoff in the table). In zero sum games, we usually will write player 1's payoff (the row player), as the other player's payoff can be inferred from the first player's payoff. When only player 1's payoff is listed, this is called a **payoff matrix**. For Rock Paper Scissors, the payoff matrix is given below.

| Payoff | Rock | Paper | Scissors |
|---|---|---|---|
| Rock | $0$ | $-1$ | $1$ |
| Paper | $1$ | $0$ | $-1$ |
| Scissors | $-1$ | $1$ | $0$ |

# Examples of Games

## Ex 1 (Prisoner's Dilemma)

One of the most famous examples of a game is Prisoner's Dilemma. Suppose COLlin (the column player) and ROb (the row player) illegally download a car. The investigating officers hope to get a confession from Collin and Rob, so they take both Collin and Rob to separate rooms so they both cannot communicate. If both people keep quiet, each of them will face one year in jail. If one speaks but the other keeps quiet, the person who speaks has no jail time, but the person who kept quiet will face five years in jail. If both speak, then both will face three years in jail. The payoffs are given in the table below:

| Payoff | Keep Quiet | Speak |
|---|---|---|
| Keep Quiet | $(-1,-1)$ | $(-5,0)$ |
| Speak | $(0,-5)$ | $(-3,-3)$ |

Even though both people have the best outcome if they both keep quiet, this isn't what game theory suggests the prisoners should do. If Collin keeps quiet, then Rob should speak. After all, Rob wants less jail time. If Collin speaks, Rob should also speak. This will give Rob less jail time. A similar argument holds for Collin. If each player acts purely in their own rational self interest, then both prisoners will have a worst combined outcome. In practice, this scenario is simpler than the real world. There are more things to consider than just years of jail time. This type of scenario models many real world dilemmas though (this has been used to explain to US and USSR arms race in the cold war).

## Ex 2 (Stag Hunt)

Suppose two hunters need to decide whether to work together to hunt a stag or to hunt for rabbits on their own. If both hunters work together, the hunters will both be able to eat for 10 days. If each hunter hunts rabbits, each hunter will be able to eat for 3 days. If one hunter attempts to hunt the stag on their own, this hunter will find 1 day of food, while the hunter who pursues the rabbit will have 4 days of food. The payoffs are summarized in the table below:

| Payoff | Stag | Rabbit |
|---|---|---|
| Stag | $(10,10)$ | $(1,4)$ |
| Rabbit | $(4,1)$ | $(3,3)$ |

Suppose Alice decides in advance to hunt for the rabbit. If she communicates this to Bob, then Bob is incentivized to hunt for rabbits as well. Bob will only stand to gain more days of food. If Alice instead decides to hunt for the stag and communicates this to Bob, then Bob is also incentivized to hunt for the stag. Both players stand to gain more by working together than against each other. As such, there are two optimal optimal strategy pairings (Stag, Stag) and (Rabbit, Rabbit).

# Analysis of Zero Sum Games

For zero sum games, there is a nice way to find the optimal strategy for each player using LPs. This method allows us to find optimal strategies even if they are mixed strategies. Suppose a game has a payoff matrix $A$. Suppose Alice's selected strategy comes from the selected row and Bob's selected strategy comes from the column. Then Bob's optimal strategy can be modeled as the LP below:

$$\text{Minimize } v$$

s.t.

$$A\vec{x} \leq v\vec{1}$$

$$x_1 + x_2 + \ldots + x_n = 1$$

$$\vec{x} \geq 0, v \in \mathbb{R}$$

Here $v$ is the expected payoff. $\vec{x}$ consists of the probability of selecting each strategy in the optimal mixed strategy. $\vec{1}$ is the vector of all ones. In particular, we have the listed inequalities $A\vec{x} \leq v\vec{1}$ since we assume that Alice is trying to maximize the payoff. This comes from the adversarial nature of the game. The dual LP gives Alice's optimal strategy.

$$\text{Maximize } w$$

s.t.

$$A^T\vec{y} \geq w\vec{1}$$

$$y_1 + y_2 + \ldots + y_m = 1$$

$$\vec{y} \geq 0, w \in \mathbb{R}$$

Once again, we have $\vec{1}$ as the vector of all 1s. We also will have $w$ as the payoff.

## Ex 3 (Rock Paper Scissors)

We will have the LPs given below for Rock Paper Scissors:

**Primal**

$$\text{Minimize } v$$

s.t.

$$\begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix} \vec{x} \leq v \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$x_1 + x_2 + x_3 = 1$$

$$\vec{x} \geq 0, v \in \mathbb{R}$$

**Dual**

$$\text{Maximize } w$$

s.t.

$$\begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \vec{y} \geq w \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$y_1 + y_2 + y_3 = 1$$

$$\vec{y} \geq 0, w \in \mathbb{R}$$

We can see both players have an optimal strategy of $\begin{pmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix}$. As these both give a value of 0 for $w$, we can see by strong duality that this strategy is optimal.

## Ex 4 (Market Share)

Suppose Alice and Bob both own marketing firms. Both are competing for the maximum market share. Suppose both companies are competing in a duopoly (so the market share is purely split between their companies). The payoff matrix (with percentages of market share) is given below:

$$A = \begin{pmatrix} 30 & 40 & 60 \\ 60 & 10 & 30 \end{pmatrix}$$

Notice the number of rows and columns need not be the same for a zero sum game. In this case, we get the following LPs for Bob and Alice respectively:

**Primal**

Minimize $v$

s.t.

$$\begin{pmatrix} 30 & 40 & 60 \\ 60 & 10 & 30 \end{pmatrix} \vec{x} \leq v \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$x_1 + x_2 + x_3 = 1$$

$$\vec{x} \geq 0, v \in \mathbb{R}$$

**Dual**

Maximize $w$

s.t.

$$\begin{pmatrix} 30 & 60 \\ 40 & 10 \\ 60 & 30 \end{pmatrix} \vec{y} \geq w \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$y_1 + y_2 = 1$$

$$\vec{y} \geq 0, w \in \mathbb{R}$$

For Bob, the optimal strategy is $\begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{pmatrix}$. For Alice the optimal strategy is $\begin{pmatrix} \frac{5}{6} \\ \frac{1}{6} \end{pmatrix}$. These yield a value of $v = w = 35$ so we see that these strategies are optimal by strong duality. They correspond to $v = w = 35$, so we see Bob's firm will have 65% of the market share.